

Changes Made from Reviewer Feedback

Document Ranking for Textbook Search
CMSC 673 Intro to NLP, Implementation Track
Paper ID: RM24810

This doc lists what I changed between the Checkpoint 2 draft and the final version. The first part is changes I made because of reviewer feedback. The second part is changes I made on my own.

Changes from reviewer feedback

- 1. Task wording.** A reviewer said the task in part (a) is that the user types a topic and the app finds relevant textbooks, not just the one chapter most likely to help. I changed Section 1 so the goal is about textbook search, and each ranked section is treated as a search result. This matches the real task better.
- 2. Number of queries.** A reviewer said 20 queries was not enough to test the system. I grew the set to 100 queries. They mix keyword, question, and comparison styles, and there are 129 graded labels in qrels.jsonl.
- 3. Labeling bias.** A reviewer said labeling my own queries can be biased. I could not get a second labeler in time, but I did two things to cut down the bias. I wrote every query before running the search even once, so I was not pushed toward what BM25 gave back. And I re-read each section before setting its label instead of going by memory. I also say plainly in the writeup that a second labeler would have been better.
- 4. Data cleaning detail.** A reviewer asked for a more detailed cleaning plan. Section 4 now walks through five steps: strip junk tags, strip junk classes and ids, split into sections at h2 and h3 headings, fix whitespace, and drop garbage sections. The same steps are in src/clean.py.
- 5. References.** A reviewer said only the main library was cited and that NLTK, the dataset, and the models were not. The references now cite rank_bm25, NLTK, sentence-transformers, the all-MiniLM-L6-v2 model card, OpenStax Biology 2e, OpenAI gpt-3.5-turbo, NumPy, Beautiful Soup, and Flask.
- 6. Dataset size.** A reviewer said the dataset is fine if I use it in full and not just a small part. The Checkpoint 2 draft used only a few passages. The final corpus has 91 sections that cover the main chapters of OpenStax Biology 2e. It is still not the whole book, and I say that in the limitations section, but it is much closer to a full corpus than the checkpoint version.

Other changes I made on my own

Score threshold for off-topic queries. Before, a query like “what is nlp” returned five biology sections with near-zero scores. I added a 0.5 score threshold and a “No relevant sections found” message so the app is honest when the corpus does not cover the topic.

More metrics. My Checkpoint 2 plan only had Precision@3 and NDCG@5. The final version also reports Recall@5 and F1@5, so the table shows the precision and recall tradeoff as well.

More sanity tests. I went from three correctness checks to seven. The new ones cover the off-topic threshold, the k parameter, and an empty query.

Web app. I added a small Flask web app so the search works in a browser, and put two screenshots of it in the writeup.